Blessed G/Michael Catholic Secondary School
1st Semester Chemistry Test One for Grade 9th
NOV, 2018 E.C
Time Allowed: 1 hr

Name: _____ **Sec:** _____ **R.No.** _____

Instruction I- write true if the statement is correct and false otherwise

_____1. Chemistry does not study the energy changes associated with chemical reactions.

_____2. Analytical chemistry is only concerned with producing chemicals, not analyzing them

_____3. Industrial processes such as brewing, soap making, and dye production rely on chemical principles.

_____4. All measurements in chemistry must include a number and a unit.

Instruction II - choose the best answer from the given alternatives

_____1. The primary focus of chemistry is the study of:
A) Living organisms and their functions
B) The motion and forces affecting objects
C) The composition, structure, properties, and changes of matter
D) The history and evolution of the universe

_____2. Which of the following best describes the "matter" that chemists study?
A) Only the substances that can be seen and touched
B) Any substance that has mass and takes up space
C) Only the elements found on the Periodic Table
D) Only complex compounds, such as plastics and medicines

_____3. A chemist who is studying the rate at which a chemical reaction occurs is working in the branch of chemistry known as:
A) Analytical chemistry
B) Inorganic chemistry
C) Physical chemistry
D) Bio chemistry

_____4. The process of observing a substance and determining its components and their relative amounts is primarily the focus of which branch of chemistry?
A) Physical chemistry
B) Analytical chemistry
C) Organic chemistry
D) Bio chemistry

_____5. A researcher is studying a newly discovered compound containing only carbon and hydrogen atoms. This falls under the branch of chemistry called:
A) Inorganic chemistry
B) Biochemistry
C) Organic chemistry
D) Analytical chemistry

_____6. In industrial production, chemistry plays a critical role in converting raw materials into useful products such as:

A) Solar panels, wind turbines, and hydropower stations
B) Plastics, medicines, and fertilizers
C) Computer software, mobile apps, and video games
D) Books, newspapers, and magazines

_____7. In agriculture, chemistry is primarily used to:

A. Create new types of soil
B. Develop fertilizers and pesticides that increase crop yield
C. Build farming machinery
D. Control weather conditions

_____8. Petrochemical industries rely on chemistry mainly to:

A. Grow crude oil naturally
B. Refine crude oil into fuels and useful chemical products
C. Remove all carbon from fuels
D. Convert metals into plastics

_____9. Which of the following best defines the **chemical industry**?

A. The industry that produces machines and mechanical tools.
B. The industry that extracts minerals from the earth.
C. The industry that converts raw materials into useful products through chemical processes.
D. The industry that provides banking and financial services.

_____10. Which of the following is **not** a major sector of the chemical industry?

A. Petrochemical industry
B. Pharmaceutical industry
C. Textile chemical processing
D. Automobile repair industry

_____11. Which of the following is an example of a **chemical product**?

A. Crude oil directly from the ground
B. Water from a river
C. Plastic made from polymerization of ethene
D. Wood cut from a tree

_____12. The **composition of matter** refers to:

A. The physical appearance of matter.
B. The types and proportions of atoms or elements that make up a substance.
C. The amount of energy matter contains.
D. The temperature of matter.

_____13. Which of the following is **not** a basic physical quantity?

A. Mass
B. Length
C. Temperature
D. Concentration

____14. **Temperature** in the SI system is measured in:

A. Celsius (°C)
B. Fahrenheit (°F)
C. Kelvin (K)
D. Joule (J)

_____15. Which of the following quantities measures the **space occupied by matter**?

A. Density
B. Volume
C. Mass
D. Pressure

_____16. Which of the following is a **derived physical quantity** in chemistry?

A. Time
B. Mass
C. Density
D. Amount of substance

_____17. A chemist measures 2.5 mol of oxygen gas. The term *mole* refers to:

A. The mass of oxygen gas
B. The number of particles present
C. The temperature of oxygen
D. The pressure of the gas

_____18. Temperature is a measure of:

A. The total energy of a system
B. The average kinetic energy of particles in a substance
C. The potential energy of atoms
D. The size of molecules

_____19. Which statement about the Kelvin scale is **true**?

A. It is based on the freezing and boiling points of water.
B. It has negative temperature values.
C. It begins at absolute zero where all molecular motion stops.
D. It is used only in meteorology.

_____20. Convert **0 °C** to **Fahrenheit**.

A. 32 °F B. 0 °F C. 100 °F D. −32 °F

_____21. If a sample of gas is heated from **27 °C to 327 °C**, by how many **Kelvin** does the temperature increase?

A. 100 K B. 200 K C. 300 K D. 600 K

_____22. At what temperature are **Celsius and Fahrenheit equal**?

A. 0 °C B. −32 °C C. −40 °C D. 40 °C

Instruction II- convert the following units

_____1. 1200 μm = _____ mm.

_____2. 1 megagram (Mg) = _____ kilograms (kg)

_____3. 4×10^{-5} ng = _____ mg

_____4. 8×10^{-3} pm = _____ μm

$F = \frac{9}{5}C + 32$

Time allotted: 1h

Chemistry Answer Sheet G. 9

Name Abdallah Soliman Sec. D No. 40

True or false

1. False
2. False
3. True
4. True

Multiple choice

1. C
2. B
3. A C
4. A B
5. C
6. B
7. B
8. B
9. C
10. D
11. C
12. B
13. D
14. C
15. B
16. C
17. D 1
18. A B
19. C
20. A
21. D C
22. D C

Short answer

1. 12×10^{-1}
2. $1000 = 1 \times 10^{3}$
3. 4×10^{-11}
4. 3×10^{-9}

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